cp_5

1.

Greater than 30 and 50 | 20/20

Easy

Write a program that reads a number `X` and checks,

- If `X` is greater than **30**.
- If `X` is greater than **30**, check if `X` is also greater than **50**.

Print **X is greater than 30** if `X` is greater than **30**.

Print **X is greater than 30** and **X is greater than 50** on each line if `X` is greater than **50**.

Input

The input will be a single line containing an integer representing `X` .

Output

If `X` is greater than **30**,

The output should be a single line containing the string **X is Greater than 30**.

If `X` is greater than **30** and if `X` is also greater than **50**,

The first line of output should be a string containing **X is Greater than 30**.

The second line of output should be a string containing **X is Greater than 50**.

Explanation

For example, if the given number is `X = 45` ,

✓ `X` is greater than **30**. (**45** is greater than **30**)

✕ `X` is greater than **50**. (**45** is **not** greater than **50**)

The output should be **X is Greater than 30**.

For example, if the given number is `X = 99` ,

✓ `X` is greater than **30**. (**99** is greater than **30**)

✓ `X` is greater than **50**. (**99** is greater than **50**)

The output should be,

```
X is Greater than 30
X is Greater than 50
```

Sample Input 1

```
45
```

Sample Output 1

```
X is Greater than 30
```

code:

```python
x=int(input())
is_x_greater_than_30=x>30
is_x_greater_than_50=x>50
if is_x_greater_than_30:
    print("X is Greater than 30")
    if is_x_greater_than_50:

        print("X is Greater than 50")
```

2.

Rankers | 25/25 ● Solved

Easy

Write a program that reads the rank `R` of a student and checks,

- If `R` is less than or equal to **3**.
- If `R` is **not** less than or equal to **3**, check if `R` is less than or equal to **10**.

Print **One of Top 3** if the `R` is less than or equal to **3**.
Print **Not Top 3 but One of Top 10** if `R` is less than or equal to **10** but **not** less than or equal to **3**.

Input

The input will be a single line containing an integer representing `R` .

Output

The output should be a single line containing a string. **One of Top 3** should be printed if `R` is less than or equal to **3**. **Not Top 3 but One of Top 10** should be printed if `R` is less than or equal to **10** but **not** less than or equal to **3**.

Explanation

For example, if the given rank is `R = 7` ,
✕ `R` is less than or equal to **3**. (**7** is **not** less than or equal to **3**)
✓ `R` is less than or equal to **10**. (**7** is less than or equal to **10**)
The output should be **Not Top 3 but One of Top 10** as **7** is less than or equal to **10** but **not** less than or equal to **3**.

Sample Input 1

```
7
```

Sample Output 1

```
Not Top 3 but One of Top 10
```

Sample Input 2

```
3
```

Sample Output 2

```
One of Top 3
```

code:

```python
rank=int(input())
is_rank_top_3=rank<=3
is_rank_top_10=rank<=10
if is_rank_top_3:
    print("One of Top 3")
else:
    if is_rank_top_10:
        print("Not Top 3 but One of Top 10")
```

3.

Weight | 20/20 ● Solved

Easy

Write a program that reads the weight `W` of a box in kg and checks,

- If `W` is greater than or equal to **100**.
- If `W` is **not** greater than or equal to **100**, check if `W` is greater than or equal to **30**.

Print **Box is Heavier** if `W` is greater than or equal to **100**.
Print **Box is Heavy** if `W` is **not** greater than or equal to **100** but greater than or equal to **30**.

Input

The input will be a single line containing an integer representing `W` .

Output

The output should be a single line containing a string. **Box is Heavier** should be printed if `W` is greater than or equal to **100**. **Box is Heavy** should be printed if `W` is **not** greater than or equal to **100** but greater than or equal to **30**.

Explanation

For example, if the given weight of the box is `W = 60` ,
× `W` is greater than or equal to **100**. (**60** is **not** greater than or equal to **100**)
✓ `W` is greater than or equal to **30**. (**60** is greater than or equal to **30**)
The output should be **Box is Heavy** as **60** is **not** greater than or equal to **100** but greater than or equal to **30**.

Sample Input 1

```
60
```

Sample Output 1

```
Box is Heavy
```

Sample Input 2

```
150
```

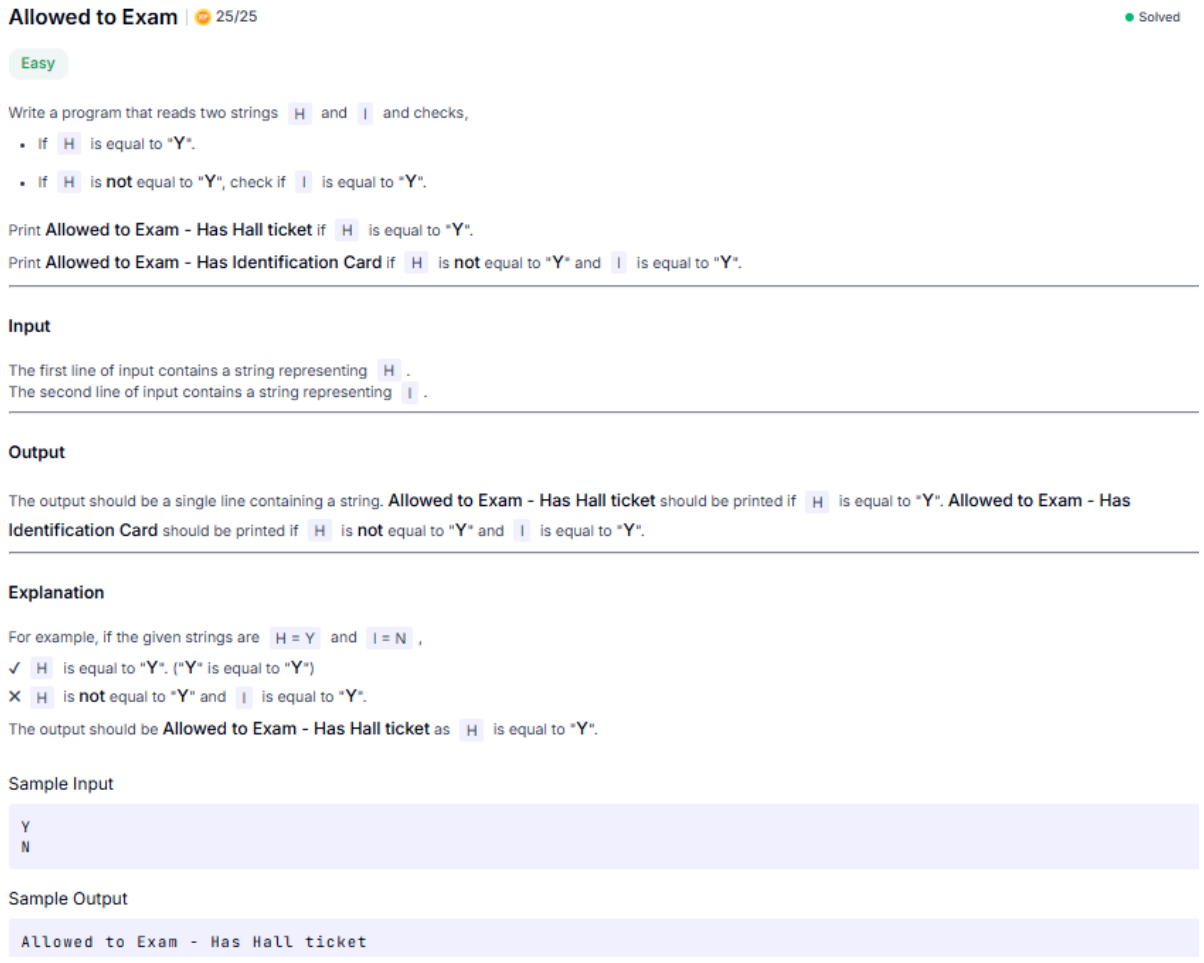
Sample Output 2

```
Box is Heavier
```

code:

```
weight=int(input())
is_box_greater_than_100=weight>=100
is_box_greater_than_30=weight>=30
if is_box_greater_than_100:
    print("Box is Heavier")
else:
    if is_box_greater_than_30:
       print("Box is Heavy")
```

4.



Allowed to Exam | 25/25 — Solved

Easy

Write a program that reads two strings `H` and `I` and checks,

- If `H` is equal to "**Y**".
- If `H` is **not** equal to "**Y**", check if `I` is equal to "**Y**".

Print **Allowed to Exam - Has Hall ticket** if `H` is equal to "**Y**".
Print **Allowed to Exam - Has Identification Card** if `H` is **not** equal to "**Y**" and `I` is equal to "**Y**".

Input

The first line of input contains a string representing `H` .
The second line of input contains a string representing `I` .

Output

The output should be a single line containing a string. **Allowed to Exam - Has Hall ticket** should be printed if `H` is equal to "**Y**". **Allowed to Exam - Has Identification Card** should be printed if `H` is **not** equal to "**Y**" and `I` is equal to "**Y**".

Explanation

For example, if the given strings are `H = Y` and `I = N` ,
✓ `H` is equal to "**Y**". ("**Y**" is equal to "**Y**")
× `H` is **not** equal to "**Y**" and `I` is equal to "**Y**".
The output should be **Allowed to Exam - Has Hall ticket** as `H` is equal to "**Y**".

Sample Input

```
Y
N
```

Sample Output

```
Allowed to Exam - Has Hall ticket
```

code:

```
h=input()
i=input()
```

```
is_h_equal_to_y=h=="Y"
is_i_equal_to_y=i=="Y"
if is_h_equal_to_y:
    print("Allowed to Exam - Has Hall ticket")
else:
    if is_i_equal_to_y:
        print("Allowed to Exam - Has Identification Card")
```

5.

Divisible by 10 or 5 | 20/20 ● Solved

Easy

Write a program that reads a number `N` and checks if `N` is divisible by **5** and **10**.
Print **Divisible by 10** if `N` is divisible by **10**.
Print **Divisible by 5** if `N` is divisible by **5** but **not** divisible by **10**.
Print **Not Divisible by 10 or 5** if `N` is **not** divisible by **10** and `N` is **not** divisible by **5**.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be a single line containing a string. **Divisible by 10** should be printed if `N` is divisible by **10**. **Divisible by 5** should be printed if `N` is divisible by **5** but **not** divisible by **10**. **Not Divisible by 10 or 5** should be printed if `N` is **not** divisible by **10** and **5**.

Explanation

For example, if the given number is `N = 15` ,
✗ `N` is divisible by **10**. (**15** is **not** divisible by **10**)
✓ `N` is divisible by **5** but **not** divisible by **10**. (**15** is divisible by **5**. **15** is **not** divisible by **10**)
The output should be **Divisible by 5** as `N` is divisible by **5** but **not** divisible by **10**.

For example, if the given number is `N = 11` ,
✗ `N` is divisible by **10**. (**11** is **not** divisible by **10**)
✗ `N` is divisible by **5** but **not** divisible by **10**. (**11** is **not** divisible by **5** and **10**)
The output should be **Not Divisible by 10 or 5** as `N` is **not** divisible by **10** and **5**.

Sample Input 1

```
15
```

Sample Output 1

```
Divisible by 5
```

Sample Input 2

```
5
```

Sample Output 2

```
Divisible by 5
```

code:

```
num=int(input())
is_divisible_by_10=num%10==0
is_divisible_by_5=num%5==0
if is_divisible_by_10:
    print("Divisible by 10")
else:
    if is_divisible_by_5:
        print("Divisible by 5")
    else:
        print("Not Divisible by 10 or 5")
```

6.

Greatest Among Three Numbers - 2 | 20/20

Easy

Write a program that reads three numbers `A` , `B` , and `C` , and prints the greatest number among the three given numbers.

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .
The third line of input contains an integer representing `C` .

Output

The output should be a single line containing an integer that the greatest number among the three given numbers.

Explanation

For example, if the given numbers are `A = 10` , `B = 15` , and `C = 20` ,

- `B` is greater than `A` . (**15** is greater than **10**)
- `B` is less than `C` . (**15** is less than **20**)
- So, `C` is greater than `A` and `B` .

The output should be **20**.

For example, if the given numbers are `A = -10` , `B = 59` , and `C = 34` ,

- `B` is greater than `A` . (**59** is greater than **-10**)
- `B` is greater than `C` . (**59** is greater than **34**)
- So, `B` is greater than `A` and `C` .

The output should be **59**.

Sample Input 1

```
10
15
20
```

Sample Output 1

```
20
```

Sample Input 2

```
-10
59
34
```

Sample Output 2

```
59
```

code:

```python
a=int(input())
b=int(input())
c=int(input())
is_a_greater=a>b and a>c
is_b_greater=b>a and b>c
if is_a_greater:
    print(a)
else:
    if is_b_greater:
        print(b)
    else:
        print(c)
```

7.

Course Fee | 25/25 — Solved

Easy

Write a program that reads the marks `M` of a student and checks,

- If `M` is greater than or equal to **90**.
- If `M` is greater than or equal to **50** but **not** greater than or equal to **90**

Print **Discount is 200** if `M` is greater than or equal to **90**.
Print **Discount is 100** if `M` is greater than or equal to **50** but **not** greater than or equal to **90**.
Print **No Discount** if `M` is **not** greater than or equal to **50**.

Input

The input will be a single line containing an integer representing `M`.

Output

The output should be a single line containing a string. **Discount is 200** should be printed if `M` is greater than or equal to **90**. **Discount is 100** should be printed if `M` is greater than or equal to **50** but **not** greater than or equal to **90**.**No discount** should be printed if `M` is **not** greater than or equal to **50**.

Explanation

For example, if the given marks `M = 93`,
✓ `M` is greater than or equal to **90**. (**93** is greater than or equal to **90**)
The output should be **Discount is 200**.

For example, if the given marks `M = 35`,
✗ `M` is greater than or equal to **90**. (**35** is **not** greater than or equal to **90**)
✗ `M` is greater than or equal to **50** but **not** greater than or equal to **90**. (**35** is **not** greater than or equal to **50**)
The output should be **No Discount** as `M` is **not** greater than or equal to **50**.

For example, if the given marks `M = 75`,
✗ `M` is greater than or equal to **90**. (**75** is **not** greater than or equal to **90**)
✓ `M` is greater than or equal to **50** but **not** greater than or equal to **90**. (**75** is greater than or equal to **50**)
The output should be **Discount is 100**.

Sample Input 1

```
93
```

Sample Output 1

```
Discount is 200
```

Sample Input 2

```
35
```

Sample Output 2

```
No Discount
```

code:

```python
marks=int(input())
is_marks_greater_than_90=marks>=90
is_marks_greater_than_50=marks>=50
if is_marks_greater_than_90:
    print("Discount is 200")
else:
    if is_marks_greater_than_50:
        print("Discount is 100")
    else:
        print("No Discount")
```

8.

Win or Lose or Draw | 25/25

Solved

Easy

Write a program that reads the two scores `A` and `B` and compares `A` with the `B`.
Print **Win** if `A` is greater than `B`.
Print **Draw** if `A` is equal to `B`.
Print **Lose** if `A` is less than `B`.

Input

The first line of input contains an integer representing `A`.
The second line of input contains an integer representing `B`.

Output

The output should be a single line containing a string. **Win** should be printed if `A` is greater than `B`. **Draw** should be printed if `A` is equal to `B`. **Lose** should be printed if `A` is less than `B`.

Explanation

For example, if the given scores are `A = 26` and `B = 47`,
✕ `A` is greater than `B`. (**26** is **not** greater than **47**)
✕ `A` is equal to `B`. (**26** is **not** equal to **47**)
✓ `A` is less than `B`. (**26** is less than **47**)
The output should be **Lose**.

Sample Input 1

```
26
47
```

Sample Output 1

```
Lose
```

Sample Input 2

```
24
15
```

Sample Output 2

```
Win
```

code:

```
a=int(input())
b=int(input())
if a>b:
    print("Win")
else:
    if a==b:
        print("Draw")
    else:
        print("Lose")
```

9.

Get Grades | 20/20

Write a program to calculate the grade of the student based on the marks he/she scored.

Input

The input will be a single line containing a float value representing `N` .

Output

The output should be a single line containing a string representing the grade of the student.

- **A** should be printed if the given marks are greater than **85**.
- **B** should be printed if the given marks are greater than **70** and less than or equal to **85**.
- **C** should be printed if the given marks are greater than or equal to **60** and less than or equal to **70**.
- **F** should be printed if the given marks are less than **60**.

Explanation

If the given input `N = 70.5` ,

✗ `N` is greater than **85**. (**70.5** is not greater than **85**)
✓ `N` is greater than **70** and less than or equal to **85**. (**70.5** is greater than **70** and less than or equal to **85**)
✗ `N` is greater than or equal to **60** and less than or equal to **70**. (**70.5** is greater than or equal to **60** and not less than or equal to **70**)
✗ `N` is less than **60**. (**70.5** is not less than **60**)

The output should be **B** as `N` is greater than **70** and less than or equal to **85**.

If the given input `N = 90` ,

✓ `N` is greater than **85**. (**90** is greater than **85**)
✗ `N` is greater than **70** and less than or equal to **85**. (**90** is greater than **70** and not less than or equal to **85**)
✗ `N` is greater than or equal to **60** and less than or equal to **70**. (**90** is greater than or equal to **60** and not less than or equal to **70**)
✗ `N` is less than **60**. (**90** is not less than **60**)

The output should be **A** as `N` is greater than **85**.

Sample Input 1

```
70.5
```

Sample Output 1

```
B
```

Sample Input 2

```
90
```

Sample Output 2

```
A
```

code:

```python
n=float(input())
if n>85:
    print("A")
elif n>=70 and n<=85:
    print("B")
elif n>60 and n<=70:
    print("C")
```

```
else:
    print("F")
```

10.

Greeting Message | 25/25

Write a program to print a greeting message based on the given time.

Input

The first line of input will be an integer.

Output

If the time is greater than or equal to 4 and less than 12, print "Good Morning".
If the time is greater than or equal to 12 and less than 16, print "Good Afternoon".
If the time is greater than or equal to 16 and less than 20, print "Good Evening".
If the time is greater than or equal to 20 or less than 4, print "Good Night".

Explanation

For example, if the given time is 9, your code should print "Good Morning".
For example, if the given time is 14, your code should print "Good Afternoon".

Sample Input 1

```
9
```

Sample Output 1

```
Good Morning
```

Sample Input 2

```
14
```

Sample Output 2

```
Good Afternoon
```

code:

```
time=int(input())
if time>=4 and time<12:
    print("Good Morning")
elif time>=12 and time<16:
    print("Good Afternoon")
```

```
elif time>=16 and time<20:
    print("Good Evening")
else:
    if time<4 or time>=20:
        print("Good Night")
```

11.

Name of the polygon | 15/15

Solved

Write a program that reads the number `N` and prints the name of the polygon based on the `N` number of sides.

Note

A polygon is a closed shape made of straight lines that has three or more sides and angles.

The below images are examples of a **Polygon** which contains three or more sides.

Input

The input will be a single line containing an integer `N` representing the number of sides.

Output

The output should be a single line containing a string representing the name of the polygon.

Not Polygon should be printed if the given number of sides is less than **3**.

Triangle should be printed if the given number of sides is equal to **3**.

Quadrilateral should be printed if the given number of sides is equal to **4**.

Pentagon should be printed if the given number of sides is equal to **5**.

Big Polygon should be printed if the given number of sides is greater than **5**.

Explanation

For example, if the given number of sides `N = 4` ,

✕ `N` is less than **3**. (**4** is not less than **3**)
✕ `N` is equal to **3**. (**4** is not equal to **3**)
✓ `N` is equal to **4**. (**4** is equal to **4**)
✕ `N` is equal to **5**. (**4** is not equal to **5**)
✕ `N` is greater than **5**. (**4** is not greater than **5**)

The output should be **Quadrilateral** as the number of sides is equal to **4**.

For example, if the given number of sides `N = 2` ,

✓ `N` is less than **3**. (**2** is less than **3**)
✕ `N` is equal to **3**. (**2** is not equal to **3**)
✕ `N` is equal to **4**. (**2** is not equal to **4**)
✕ `N` is equal to **5**. (**2** is not equal to **5**)
✕ `N` is greater than **5**. (**2** is not greater than **5**)

The output should be **Not Polygon** as the number of sides is less than **3**.

Sample Input 1

```
4
```

Sample Output 1

```
Quadrilateral
```

code:

```
sides=int(input())
if sides < 3:
    print("Not Polygon")
elif sides==3:
    print("Triangle")
elif sides==4:
    print("Quadrilateral")
elif sides==5:
    print("Pentagon")
else:
    print("Big Polygon")
```

12.

Leap Year | 25/25

Solved

Hard

Write a program that reads a year `Y` and checks if the year `Y` is a leap year. A year is a leap year if any of the given conditions are satisfied.

- `Y` is divisible by **400**.
- `Y` is divisible by **4**, and not divisible by **100**.

Print **True** if any of the given conditions are satisfied. Otherwise, print **False**.

Input

The input will be a single line containing an integer representing `Y` .

Output

The output should be a single line containing a boolean value. **True** should be printed if any of the given conditions are satisfied. Otherwise, **False** should be printed.

Explanation

For example, if the given `Y = 2016` ,

✕ `Y` is divisible by **400**. (The remainder is **16** when `2016` is divided by `400`)

✓ `Y` is divisible by **4** and not divisible by **100**. (The remainder is **0** when `2016` is divided by `4` and the remainder is **16** when `2016` is divided by `100`)

As one of the given conditions is satisfied,

The output should be **True**.

For example, if the given `Y = 1800` ,

✕ `Y` is divisible by **400**. (The remainder is **200** when `1800` is divided by `400`)

✕ `Y` is divisible by **4** and not divisible by **100**. (The remainder is **0** when `1800` is divided by `4` and the remainder is **0** when `1800` is divided by `100`)

As both the given conditions are not satisfied,

The output should be **False**.

Sample Input 1

```
2016
```

Sample Output 1

```
True
```

Sample Input 2

```
1800
```

Sample Output 2

```
False
```

code:

```python
year=int(input())
cond1=year%400==0
cond2=(year%100!=0) and (year%4==0)
if cond1 or cond2:
    print("True")
else:
    print("False")
```

13.

Relation Between Two Numbers | 20/20

Solved

Write a program that reads two numbers `A` , and `B` , and prints the relation between `A` and `B` by checking if any of the given conditions is satisfied.

- **A** is equal to **B**. (**A == B**)
- **A** is greater than **B**. (**A > B**)
- **A** is less than **B**. (**A < B**)

Note

Relational Operators are `==` , `>` , `<`

Input

The first line of input contains an integer representing `A` .
The second line of input contains an integer representing `B` .

Output

The output should be a single line containing a string representing the relation between the two numbers.

A == B should be printed if `A` is equal to `B` .

A > B should be printed if `A` is greater than `B` .

A < B should be printed if `A` is less than `B` .

Explanation

For Example, if the given integers are `A = 3` and `B = 4` ,

✕ **A** is equal to **B**. (**3** is not equal to **4**)
✕ **A** is greater than **B**. (**3** is not greater than **4**)
✓ **A** is less than **B**. (**3** is less than **4**)

The output should be **A < B** as `A` is less than `B` .

For Example, if the given integers are `A = 4` and `B = 4` ,

✓ **A** is equal to **B**. (**4** is equal to **4**)
✕ **A** is greater than **B**. (**4** is not greater than **4**)
✕ **A** is less than **B**. (**4** is not less than **4**)

The output should be **A == B** as `A` is equal to `B` .

Sample Input 1

```
3
4
```

Sample Output 1

```
A < B
```

code:

```
a=int(input())
b=int(input())
if a>b:
    print("A > B")
elif a<b:
    print("A < B")
else:
    print("A == B")
```

14.

Electricity Bill | 40/40

Solved

In this problem, you need to write a program to calculate the electricity bill for a household, based on the units of electricity the household consumed. The price for unit varies based on the slab. The charges per unit for different slabs are as mentioned below:
For the first 50 units (0-50), the charge is 2/unit
For the next 100 units (51-150), the charge is 3/unit
For the next 100 units (151-250), the charge is 5/unit
For above 250 units (251 and above), the charge is 8/unit
Apart from these charges, there is also an additional surcharge of 20% on the total amount is added to the bill.

Input

The input will be a single line containing an integer.

Output

The output should be a single line containing the calculated bill amount.

Explanation

For example, if the given number of units is 50.

```
Charges 2/unit for 0 to 50 units   ⇒  50 x 2   = 100
Charges 3/unit for 51 to 150 units ⇒   0 x 3   =   0
Charges 5/unit for 151 to 250      ⇒   0 x 5   =   0
Charges 8/unit for 251 and above   ⇒   0 x 8   =   0
-------------------------------------------------------
Total                              ⇒             100
Surcharge (20 % of Bill)           ⇒ 100 x 0.2 =  20
-------------------------------------------------------
Bill                               ⇒             120
-------------------------------------------------------
```

So the total bill should be 120.0

For example, if the given number of units is 200.

```
Charges 2/unit for 0 to 50 units   ⇒  50 x 2   = 100
Charges 3/unit for 51 to 150 units ⇒ 100 x 3   = 300
Charges 5/unit for 151 to 250      ⇒  50 x 5   = 250
Charges 8/unit for 251 and above   ⇒   0 x 8   =   0
-------------------------------------------------------
Total                              ⇒             650
Surcharge (20 % of Bill)           ⇒ 650 x 0.2 = 130
-------------------------------------------------------
Bill                               ⇒             780
-------------------------------------------------------
```

So the total bill should be 780.0

Sample Input 1

```
50
```

Sample Output 1

```
120.0
```

code:

```
units=int(input())
first_50=0
next_150=0
after_150_100=0
above_250=0
total_bill=0
```

```
additional=0
total_charge=0
if units<=50:
    first_50=units*2
elif units>50 and units<=150:
    first_50=50*2
    next_150=(units-50)*3
elif units>150 and units<=250:
    first_50=50*2
    next_150=100*3
    after_150_100=(units-150)*5
elif units >250:
    first_50=50*2
    next_150=100*3
    after_150_100=100*5
    above_250=(units-250)*8
total_charge=first_50+next_150+after_150_100+above_250
additional=total_charge*0.2
total_bill=additional+total_charge
print(total_bill)
```